

Problem Set 2

Sumedha Ray

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Main Tools of a Data Scientist

- Measurement - Understanding what data captures and doesn't capture; quantifying qualitative phenomena
- Statistical Programming Languages such as R (with tidyverse), Python, Julia, Stata, SAS, SPSS and Matlab
- Visualization Tools using ggplot2 (R), matplotlib (Python), Plots.jl (Julia) and Tableau
- Web Scraping using APIs (Application Program Interfaces) and HTML parsing (rvest, BeautifulSoup, Gumbo.jl)
- Big Data Management using SQL (Structured Query Language), Spark, Hadoop and RDD (Resilient Distributed Datasets)
- Statistical Modeling - This involves testing theories with data and explaining their behavior.